The title (*which should really have more than two words, but should not be longer than two lines*)

A. Student (*remember to remove ID for the first draft and peer review!*)

October 18, 2026

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

1 Introduction

The Domain Name System (DNS) is a fundamental component of the Internet, responsible for translating human-readable domain names into IP addresses (Mockapetris, 1987). Most interactions with the internet require DNS resolution, which makes DNS queries a source of information for users' online activity. Originally, DNS queries have been transmitted without encryption, allowing all intermediaries with access to the packet to observe which domains users access (Wicinski, 2021). Protocols such as DNS over HTTPS (DoH), DNS over TLS (DoT) and DNS over QUIC (DoQ) have been invented to provide a solution for this issue by encrypting the communication between clients and DNS servers (Hoffman and McManus, 2018; Hu et al., 2016; Huitema et al., 2022). However, while these protocols protect DNS queries from intermediaries in the network, the DNS server can still associate a query to the IP Address of the user that issued them.

Oblivious DNS over HTTPS (ODoH) addresses this issue by introducing an intermediary proxy that separates the client's identity from the query contents, forwarding encrypted requests to the target DNS server, which ensures that neither the proxy or the DNS server possess both the user's identity and the query contents (Singanamalla et al., 2020; Kinnear et al., 2022). Although this approach guarantees the privacy of the client's request end to end, it also prevents intermediary proxies from caching requests, since the query contents only get decrypted at the target resolver.

To overcome this issue, Niroula et al. (2026) developed Cacheable Oblivious DNS over HTTPS (CODoH) protocol. Rather than simply decrypting and caching DNS queries, CODoH combines trusted execution environments with techniques to prevent cache access patterns from revealing information. Its objective is to preserve the privacy properties of ODoH while partly recovering the performance benefits of DNS caching. However, these mechanisms also introduce overhead, raising questions about the performance of CODoH.

This report examines the architecture and privacy-enhancing mechanisms of CODoH.

2 Background

2.1 DNS & DNS Caching

The *Domain Name System (DNS)* is a distributed naming system that allows mapping domain names into certain information, usually into IP addresses (Mockapetris, 1987). When a client needs to resolve a domain address, usually it sends a query to a recursive *DNS resolver*, which may contact multiple DNS servers

following the DNS hierarchy, as illustrated in Figure 1. For example, a client may request the DNS resolver the resolution of the domain "example.com". In such case, the DNS resolver will initially query the Root Server to get the Top Level Domain (TLD) server information. Then, it will query the TLD server ".com" to obtain the authoritative Name Server (NS) of "example.com". Finally, it will query the NS to obtain the corresponding IP Address, and answer the original client.

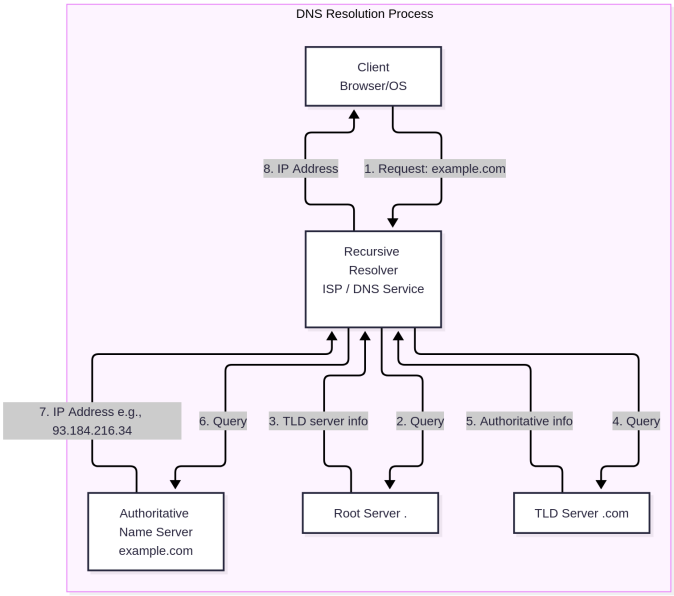


Figure 1: DNS resolution process. Initially generated using Gemini 4 Argon with the prompt “Generate a high-level mermaid diagram illustrating DNS resolution with a recursive resolver” on 17.10.2026, modified by the author.

To reduce the overhead associated with the DNS resolution process, recursive resolvers may maintain caches of previously retrieved DNS records, with records having an associated Time To Live (TTL), which determines for how long the DNS server can deliver the stored information until it is considered stale and discarded. (Mockapetris, 1987) When a resolver receives a query, it checks whether there is a valid cached answer, which would result in a *cache hit*. Otherwise, a *cache miss* happens, which forces the resolve to query the authoritative servers again. This process is illustrated in Figure 2

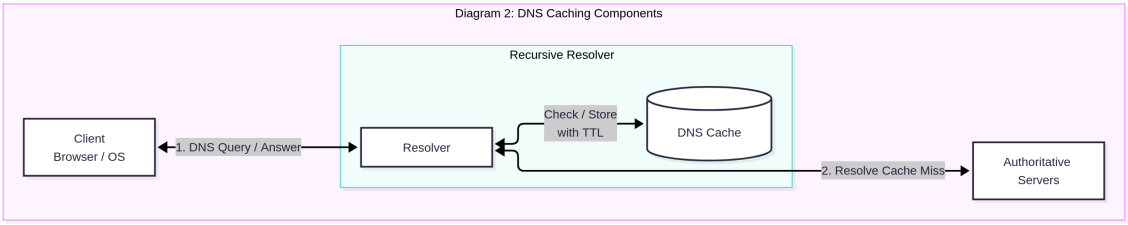


Figure 2: DNS caching. Initially generated using Gemini 4 Argon with the prompt “Generate a high-level mermaid diagram illustrating DNS recursive resolver caching, with its components and interactions” on 17.10.2026, modified by the author.

2.2 DNS Privacy: DoH & ODoH

Traditionally, DNS messages have been transported without encryption, usually encapsulated over User Datagram Protocol (UDP) (Mockapetris, 1987). This implies that network observers can read all the packet information, including Source IP Address and the requested domain name, potentially exposing browsing activity. Furthermore, the lack of encryption in traditional DNS increases its vulnerability to network at-

tacks. An attacker may intercept a query and try to manipulate the response to redirect the victim to a malicious location (Atkins and Austein, 2004).

Hoffman and McManus (2018) solved the lack of encryption by transporting DNS over HTTPS, using Transport Layer Security (TLS) for encryption, integrity and confidentiality. This is done by encapsulating DNS messages in HTTP requests under HTTPS URIs, using either POST or GET methods. DoH prevents network observers and/or attackers from reading the DNS queries, partly protecting users' privacy. However, it still has a key limitation: the resolver has access to the unencrypted DNS message and can associate it with the client's IP address.

To solve this problem, Singanamalla et al. (2020) introduced *Oblivious DNS over HTTPS (ODoH)*, which introduces a proxy between the client and the resolver. The client encrypts the DNS message for the resolver, so the proxy does not have access to the contents of the query, and forwards the encrypted DNS message, as illustrated in Figure 3. Assuming the proxy and the resolver do not collude, the client's IP address is unknown for the resolver, and the queried domain name is unknown for the proxy, so the users' privacy remains protected. However, this introduces a new problem: encryption makes two equal DNS queries different to the proxy, so conventional caching becomes impossible.

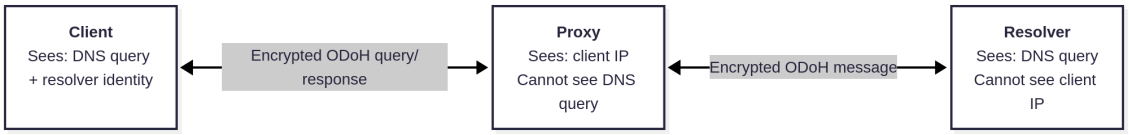


Figure 3: ODoH Architecture. Generated using GPT 5.6 Sol High with the prompt "Create a simple mermaid LR diagram to illustrate ODoH: Client <-> Proxy <-> Resolver. Show what's visible to each party" on 17.10.2026

2.3 Trusted Execution Environments (TEEs) & Remote Attestation

Although encryption protects sensitive information in transit and rest, processing this information usually requires it to be decrypted and loaded into memory. Traditionally, privileged software such as the Operating System has been allowed to access and manipulate all computer memory. Consequently, applications must trust the environment in which they execute. *Trusted Execution Environments (TEEs)* address this limitation by providing hardware-enforced isolation. (Sabt et al., 2015)

Intel SGX (Costan and Devadas, 2016) is a TEE that extends the Intel architecture to guarantee isolated execution in protected containers called *enclaves*, and permits proving to a remote computer that it is communicating with a specific secure container using *software attestation*. The proof is an attestation signature covering the container's initial state, a challenge random number produced by the remote computer, and a message produced by the container. This allows remote hosts to execute programs in a secure container, as illustrated in Figure 4.

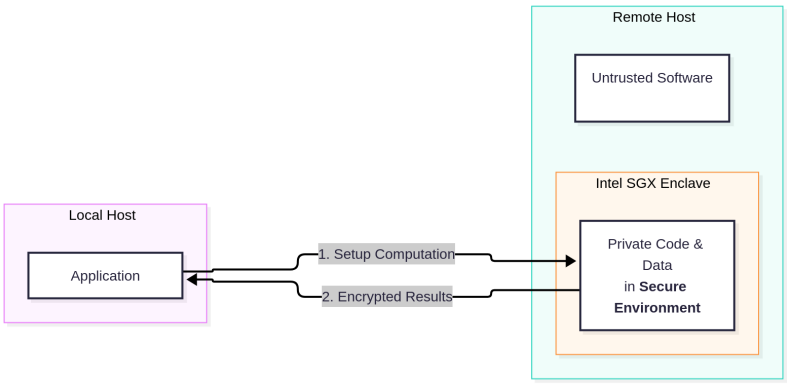


Figure 4: Intel SGX Secure Remote Computation.

2.4 Oblivious RAM (ORAM)

2.5 Related work

Here, you include short descriptions of similar work and explain *how* they are different from the main content you have presented.

Level of detail similar to an abstract.

3 Main content

4 Another topic in the main content

5 Conclusion

6 Use of generative digital tools

Large Language Models (LLMs), specifically Gemini and ChatGPT models, have been used in the creation of this report for the following purposes:

A. Research and understanding concepts, using prompts of the type: "How does [...] work?", "Explain [...]", "What solutions similar to [...] have been used to solve [...] problem?"

B. Generation of diagrams. All prompts were of the type: "Generate a mermaid diagram illustrating [...]"

C. Concept checking. Prompts of the type: "Is [...] conceptually correct?", "Is [...] correctly expressed?"

References

Atkins, D. and Austein, R. (2004). Threat Analysis of the Domain Name System (DNS). RFC 3833.

Costan, V. and Devadas, S. (2016). Intel SGX explained. Cryptology ePrint Archive, Paper 2016/086.

Hoffman, P. and McManus, P. (2018). DNS Queries over HTTPS (DoH). RFC 8484.

Hu, Z., Zhu, L., Heidemann, J., Mankin, A., Wessels, D., and Hoffman, P. (2016). Specification for DNS over Transport Layer Security (TLS). RFC 7858.

Huitema, C., Dickinson, S., and Mankin, A. (2022). DNS over Dedicated QUIC Connections. RFC 9250.

Kinnear, E., McManus, P., Pauly, T., Verma, T., and Wood, C. (2022). Oblivious DNS over HTTPS. RFC 9230.

Mockapetris, P. (1987). Domain names - concepts and facilities. RFC 1034.

Niroula, P., Gloudemans, L., Poudel, A., MacDonald, C., and Herwig, S. (2026). Codoh: Privacy-preserving caching for oblivious dns over https. *Proceedings on Privacy Enhancing Technologies*.

Sabt, M., Achemlal, M., and Bouabdallah, A. (2015). Trusted execution environment: What it is, and what it is not. In *2015 IEEE Trustcom/BigDataSE/ISPA*, volume 1, pages 57–64.

Singanamalla, S., Chunhapanya, S., Vavruša, M., Verma, T., Wu, P., Fayed, M., Heimerl, K., Sullivan, N., and Wood, C. (2020). Oblivious dns over https (odoh): A practical privacy enhancement to dns. *arXiv preprint arXiv:2011.10121*.

Wicinski, T. (2021). DNS Privacy Considerations. RFC 9076.

University of Basel
Faculty of Science

Declaration on Scientific Integrity
(including a Declaration on Plagiarism and Fraud)
Translation from German original

Title of Thesis:

Name Assessor: _____

Name Student: _____

Matriculation No.: _____

I attest with my signature that I have written this work independently and without outside help. I also attest that the information concerning the sources used in this work is true and complete in every respect. All sources that have been quoted or paraphrased have been marked accordingly.

Additionally, I affirm that any text passages written with the help of AI-supported technology are marked as such, including a reference to the AI-supported program used. This paper may be checked for plagiarism and use of AI-supported technology using the appropriate software. I understand that unethical conduct may lead to a grade of 1 or "fail" or expulsion from the study program.

Place, Date: _____ Student: _____

Will this work, or parts of it, be published?

No

Yes. With my signature I confirm that I agree to a publication of the work (print/digital) in the library, on the research database of the University of Basel and/or on the document server of the department. Likewise, I agree to the bibliographic reference in the catalog SLSP (Swiss Library Service Platform). (cross out as applicable)

Publication as of: _____

Place, Date: _____ Student: _____

Place, Date: _____ Assessor: _____

Please enclose a completed and signed copy of this declaration in your Bachelor's or Master's thesis.

September 2023